

Khalid Alghanemy

Data Engineer

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Saudi Arabia

[LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Computer Engineering graduate (GPA 4.11/5) specializing in end-to-end data pipeline design — real-time ingestion, transformation, and cloud data warehousing — using Kafka, PySpark, Airflow, and AWS (S3, Redshift). Experienced in ETL/ELT design, data modeling, and data quality validation, with a track record of turning pipeline output into stakeholder-facing dashboards.

PROJECTS

[Real-Time Market Data Pipeline](#) 2026

Personal Project

- o Built a unified Kafka producer layer to continuously ingest live crypto and stock data from Binance WebSocket and Alpha Vantage APIs into PySpark Structured Streaming.
- o Applied data modeling, cleaning, and validation logic in the PySpark transformation layer to catch inconsistent formats across multiple source systems before they reached storage.
- o Structured and loaded transformed streaming data into a dual-region cloud data warehouse, landing Parquet-formatted data in S3 (eu-north-1) and Redshift Serverless (us-east-1) for analytics and reporting.
- o Integrated Kafka/PySpark Structured Streaming with S3/Redshift storage and a Streamlit dashboard into one fully operational pipeline running end-to-end from live ingestion to visualization.

[EdgeGuard IDS – ML-Based Intrusion Detection & Risk Assessment System](#) 2025 – 2026

Graduation Project | King Khalid University

- o Built and evaluated a Random Forest classifier in scikit-learn, trained on 2.8M labeled flow records with 78 engineered features, to classify network traffic into 5 categories (BENIGN, Brute Force, DoS, PortScan, Web Attack).
- o Architected a hybrid detection pipeline layering Snort 3 signature rules with the trained ML classifier to close detection blind spots left by either approach alone.
- o Built an automated evaluation pipeline benchmarking classification results against labeled CICIDS2017 test flows to quantify model reliability per attack category.
- o Designed a scoring engine that unified ML predictions and rule-based alerts into a single 0–100 risk score across 5 severity tiers, surfaced through a live Flask dashboard with REST API and auto-refresh.

CERTIFICATIONS

- o IBM Data Engineering Professional Certificate
- o Microsoft PL-300: Power BI Data Analyst
- o [KAUST AI Certificate — Passed 2 Levels](#) | Nov 2025 – Dec 2025
- o [Saudi Council of Engineers \(SCE\)](#) — Accredited, Computer Engineering

WORK EXPERIENCE

Trainee – Subscriber Management Department

Jun 2024

Saudi Electricity Company – Abha, Saudi Arabia

- o Shadowed the Subscriber Management department for one week, documenting the coordination workflow between central and branch offices to build a working understanding of enterprise issue-resolution processes.

EDUCATION

[Bachelor's Degree in Computer Engineering](#)

2021 – 2026

King Khalid University | GPA: 4.11 / 5.00

TECHNICAL SKILLS

Data Engineering: Apache Kafka, PySpark, Apache Airflow, ETL/ELT Design, Data Modeling, Data Warehousing

Cloud & Infrastructure: AWS (S3, Redshift), Docker

Data Quality & Governance: Data Validation, Anomaly Detection, Schema Design

Programming & Databases: Python (Pandas, NumPy), SQL (MySQL, PostgreSQL), Git

Visualization & BI: Power BI (DAX, Power Query — pursuing PL-300), Streamlit, Flask (REST APIs)

Machine Learning: scikit-learn, PyTorch

LANGUAGES

Arabic: Native Speaker

English: [IELTS Academic — Overall Band 6.0 — Jul 2026](#)